

To what extent were Brexit advocacy and unemployment the definitive factors behind the conservative party's 2019 electoral success?

The 2019 United Kingdom's general election resulted in a parliamentary majority vote for the Conservative Party, fundamentally reshaping the British political landscape by enabling its departure from the European Union (EU), especially after the 2016 EU referendum (Prosser, 2020). This electoral outcome sparked debates about significant political realignment within British voting behavior, prompting curiosity over its underlining drivers. This research paper aims to contribute to this political debate by conducting quantitative constituency-level analysis to identify the socio-economic and political factors that defined the conservative party's winning vote.

This analysis is based on the 2019 British Election Study [BES]. This dataset offers comprehensive electoral details of constituencies within the UK. It spans over 14 years of socio-economic data from the year 2005 to 2019, combined with socio-demographic data from the 2011 Census (Fieldhouse et al., 2022). The data frame for this study will contain 631 cases of parliamentary constituencies, after handling missing data to ensure a complete and substantial analysis.

Our research is based off at a constituency level, and not on individual voters. Therefore, the unit of analysis is the parliamentary constituency, denoted as ***ConstituencyName***. Additionally, the unit of measurement for all variables in this paper are percentage points, ranging from (0% - 100%). Our levels of measurement are in continuous-ratio for each variable.

The variables include:

1. **Con19**: This represents the 2019 Conservative Vote Share. It contextualizes the electoral outcome we aim to explain; it will be represented in our analysis as the ***dependent variable (Y)***.
2. **LeaveHanretty**: This is the estimated "leave" vote share of the percentage of voters in each constituency; it will be denoted as an ***independent variable (X₁)***.
3. **C11Unemployed**: This variable that depicts the unemployed percentile rate of each constituency from the 2011 census. This is our second ***independent variable (X₂)***.

This analysis investigates a possible two-dimensional empirical relationship. It seeks to determine how strongly related constituency-level Brexit advocacy and unemployment rate impacted the 2019 electoral success. Historically, economic insecurity caused by

unemployment can be drivers of political behavior (Bossert et al., 2022, p.803).

Additionally, we expect that within the speculation of Brexit debate, constituencies with a higher unemployment rate would appeal to the conservative political populist message. By examining these factors together, this research paper assesses whether the Conservative 2019 electoral victory was guaranteed by a coalition of these socio-economic factors, **LeaveHanretty** and **C11Unemployment**.

Null Hypothesis (H_0): Brexit advocacy and the unemployment rate did not have a linear relationship with the conservative vote share of 2019.

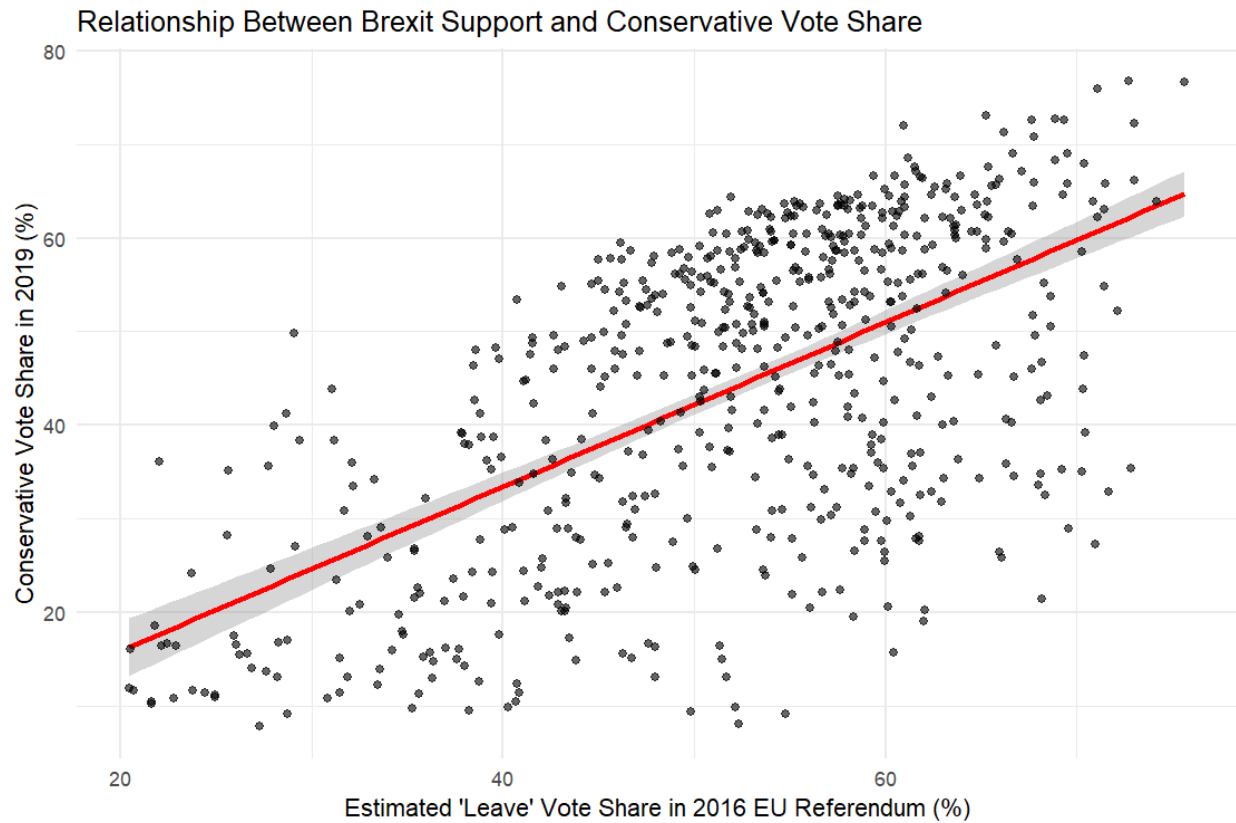
Alternative Hypothesis (H_1): Conservative vote share had a positive association with both Brexit support and unemployment rates.

Firstly, we will calculate the strength of a relationship between the conservative vote share and Brexit support. Our goal is to quantify the association between these factors with a bivariate regression model that will evaluate our hypothesis. Theoretically, we expect that constituencies with stronger patriotic support for Brexit to have higher vote shares for the Conservative Party. This expectation is grounded in the ideology that a sense of socio-economic alienation and political marginalization (McNeil and Haberstroh, 2022) in many constituencies became a driving factor of support for the Conservatives' "Get Brexit Done" campaign. Consequently, we presume that it was almost inevitable that constituencies with such sentiment would be more inclined to vote Conservative.

Null Hypothesis(H_0): There is no linear relationship between Brexit support and Conservative vote share ($\beta_1 = 0$).

Alternative Hypothesis(H_1): There is a positive linear relationship between Brexit support and Conservative vote share ($\beta_1 > 0$).

Figure 1



Advocacy for Brexit and Conservative Vote share

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Dependent variable:

Conservative Vote Share of 2019

2016 Brexit support vote share(%)	0.878*** (0.047)
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Constant	-1.671 (2.483)
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Observations	631
R2	0.361
Adjusted R2	0.360

Residual Std. Error 13.385 (df = 629)

F Statistic 354.808*** (df = 1; 629)

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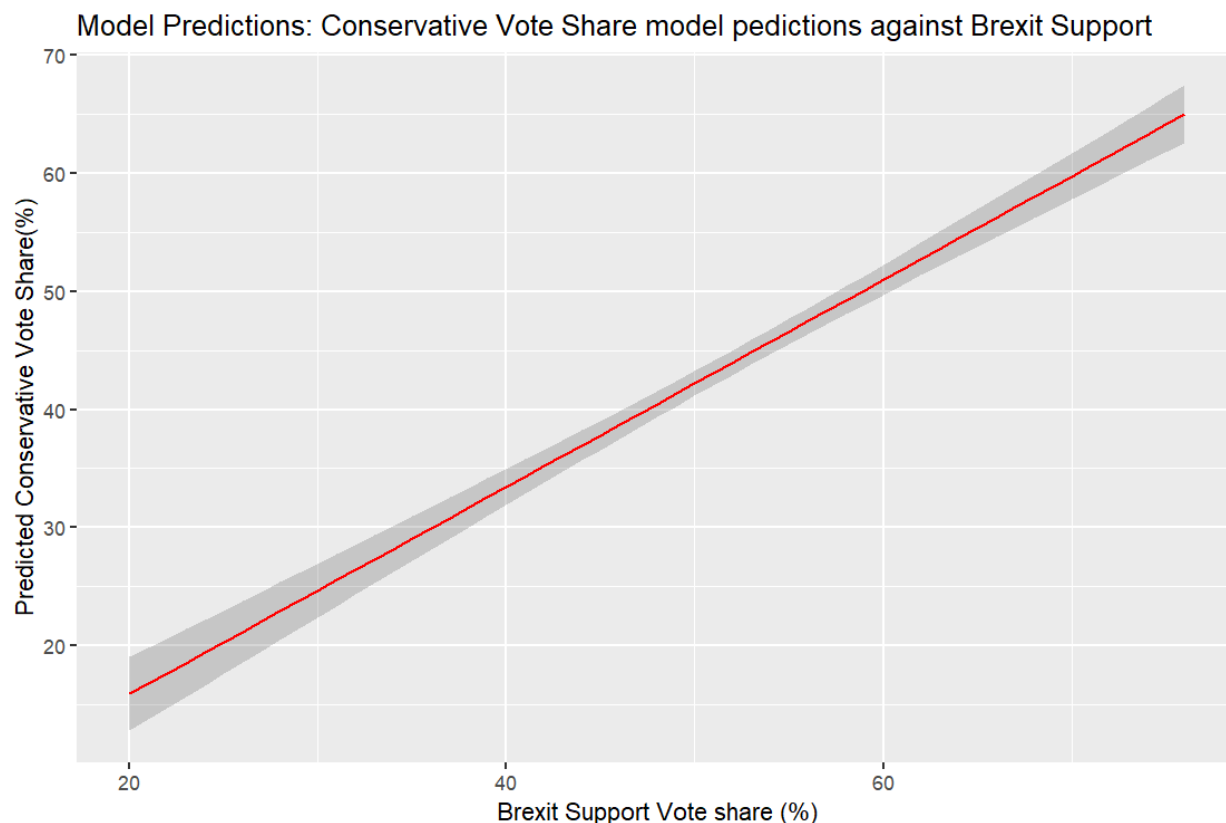
Note: *p<0.1; **p<0.05; ***p<0.01

From the image above, **Figure 1**, we constructed a bivariate analysis that reveals the relationship that provoked this research study. *How much of a significant factor was the vote to leave the EU (LeaveHanretty), a key driver to the conservative party's electoral win (Con19)?* The regression output presents a clear positive linear relationship with an upward direction. Additionally, our coefficient table explains the qualifiable significance of this relationship—indicating that “a one-percentage point increase in a vote for Brexit, is associated with an approximate 0.88 percentage points increase in conservative vote share ($\beta = 0.878$, $p < 0.001$). Our results show that this relationship is statistically significant. Therefore, constituencies with higher sentiment towards Brexit are 0.88% more likely to vote for the conservative party.

Substantially, our model explains 36% of the variance ($R^2 = 0.361$) in the conservative vote share, establishing Brexit advocacy as a fundamental driving motivator in the 2019 Conservative victory. With these results, our model enables us to reject our null hypothesis ($H_0: \beta = 0$), that depicts a negative association between Brexit support vote and the conservative vote share within British constituencies.

Moreover, to access the accuracy of our model for linearity assumptions, we will plot the prediction values of our dependent variable, **Con19**, against the values of our independent variable, **LeaveHanretty**.

Figure 2.



The plotted prediction graph in Figure 2 shows the best fit for the line on this model. It also confirms the linear relationship identified in Figure 1. For instance, in a constituency with a 45% support vote towards Brexit, the model predicts a 37.8% Conservative vote share at a 95% confidence interval of (37% - 39.1%). The model predictions remain precise across most values, confirming a consistent upward trend.

Furthermore, Model1 quantified the association between Brexit support and Conservative Vote share, indicating that Brexit Advocacy alone explains approximately 36% of the variance in the Conservative party's success. While this remains substantial, it still leaves a majority of the variation unexplained. And to fully address our research question, we expand our analysis. We will introduce the constituency-level unemployment rate into our second model, testing whether a constituency's economic vulnerability alongside Brexit support can provide a more complete explanation of the Conservative electoral outcome.

Specifically, Model 2 will address the second aspect of our research question: whether unemployment constitutes a definitive factor in Conservative success alongside Brexit?

Hypothesis for Brexit advocacy (LeaveHanretty) and Conservative vote share (Con19) — controlling for unemployment.

Null Hypothesis (H_0): Brexit support had no effect on Conservative vote, while holding unemployment constant ($\beta_1 = 0$).

Alternative Hypothesis (H_1): Brexit support had a positive effect on conservative vote share, while holding unemployment constant ($\beta_1 > 0$).

Hypothesis for Unemployment (c11Unemployment) and Conservative vote share (Con19) — controlling for Brexit.

Null Hypothesis (H_0): Unemployment had no effect on Conservative vote, while holding Brexit support constant ($\beta_2 = 0$).

Alternative Hypothesis (H_1): Unemployment had an effect on conservative vote share, while holding Brexit support constant ($\beta_2 \neq 0$).

Brexit support and Unemployment rate on Conservative vote share in 2019

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Dependent variable:

Conservative Vote Share of 2019

2016 Brexit support Vote share(%)	0.945*** (0.031)
Unemployment Rate	-7.083*** (0.249)
Constant	25.815*** (1.904)

Observations	631
R2	0.721
Adjusted R2	0.720
Residual Std. Error	8.847 (df = 628)
F Statistic	812.113*** (df = 2; 628)

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Note: *p<0.1; **p<0.05; ***p<0.01

Our second regression model stipulates that both support for Brexit and unemployment were significant predictors of Conservative vote share, though operating in opposite direction. A one percentage point increase in a constituency's "leave" vote share, is associated with a 0.945 percentage point increase in Conservative Support ($\beta = 0.945$, $p < 0.001$), holding unemployment constant. It confirms Brexit advocacy as a strong positive driver of the party's victory.

Although, a percentage point increase in local unemployment rate is associated with a 7.083 percentage point decrease in Conservative vote share ($\beta = -7.083$, $p < 0.001$), holding Brexit support constant. This negative relationship contradicts our initial hypothesis that the unemployment rate would boost Conservative support. We reject the null hypothesis of ($H_0: \beta = 0$) because the results we found show the coefficient is still statistically significant ($p < 0.001$). However, we discovered the direction of the effect is negative, unlike the positive association we expected. While unemployment is confirmed as a statistically significant factor, it reduced rather than increased Conservative vote share.

Substantively, Brexit support and unemployment rate explains 72.1% of the variance ($R^2 = 0.721$) in the Conservative electoral success, indicating that a combination of these variables represents the key factors that delivered the electoral outcome.

Comparison table for model 1 and model 2 estimates

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Dependent variable:		

Conservative Vote Share of 2019		
Model 1: Brexit Only	Model 2: Brexit + Unemployment Rate	
(1)	(2)	

2016 Brexit 'leave' Vote share(%)	0.878*** (0.047)	0.945*** (0.031)
Unemployment Rate		-7.083*** (0.249)

Constant	-1.671 (2.483)	25.815*** (1.904)

Observations	631	631
R2	0.361	0.721
Adjusted R2	0.360	0.720
Residual Std. Error	13.385 (df = 629)	8.847 (df = 628)
F Statistic	354.808*** (df = 1; 629)	812.113*** (df = 2; 628)
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Note:	*p<0.1; **p<0.05; ***p<0.01	

In this stage of the analysis, we compare both our models to assess how accounting for the unemployment rate alters our understanding of the Brexit effect and improves our quantification. The coefficient for Brexit support increases from 0.878 to 0.945 when unemployment is controlled for, suggesting its effect was slightly underestimated in the bivariate analysis due to omitted variable bias. Additionally, our R-squared doubles and rises from 36% (0.361) to 72% (0.721), meaning that Model 2 explains twice the proportion of variance within the Conservative vote share. These results were confirmed by an ANOVA test ($F = 811.97$, $p < 0.001$), which proves that adding unemployment significantly improves our model fit. The Residual Sum of Squares (RSS) drops from 112,699 to 49,150, indicating Model 2 reduces unexplained variance by 63,549 units. Overall, our comparison demonstrates that both patriotic and economic factors shaped the 2019 electoral performance. The Conservatives' victory emerged from a coalition where strong Brexit support in constituencies with low unemployment rate overcame weaker performances in areas with higher unemployment rates. While these factors in Model 2 explain 72% of the variance, the remaining 28% suggest room for other influences, such as education levels or religious beliefs, in a more comprehensive account.

However, our research acknowledges several limitations that warrant consideration. **First**, the unemployment rate for each constituency originates from the 2011 Census, while the election was held in 2019; local unemployment levels may have shifted in the years between. **Second**, our analysis is conducted at a constituency-level, preventing individual-level statistical hypothesis about voting behaviors. **Third**, there is a potential for omitted variable bias—as many factors such as campaign spending—could influence the estimate.

This paper demonstrated that both constituency-level unemployment rate and Brexit support were definitive factors in the Conservative Party's 2019 electoral success, while

impacting in opposite directions. Simultaneously, these factors explain 72.1% of variation in constituency-level support towards the Conservative victory. These findings, though with its limitations, reveal an election shaped by both patriotic sentiment and persistent economic voting that delivered the conservative majority.

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